SDG 3: Good Health and Well-Being – Health Tracking System

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I. INTRODUCTION

Background of the Study

The Health Tracker System has been designed to give the user a digital solution to allow individuals to monitor their body measurements, record symptoms, and assess their health digitally within a secure online framework. Users can create personal logins, securely access them, record their Body Mass Index (BMI), log their symptoms, and view their historical health records all via a computer-based system that allows users to easily track and manage their health data in an organized and timely manner.

The Health Tracker System has been developed using the VB.NET Windows Forms Application Development Environment integrated with Microsoft SQL Server LocalDB via ADO.NET SqlClient for secure database connectivity and data persistence.

Problem Statement

The study aims to solve the following problems:

1. Difficulty in manually recording and monitoring health information.
2. Lack of organized storage for BMI and health assessment records.
3. Difficulty in retrieving previous health monitoring data.
4. Inaccurate recording caused by manual computations.
5. Lack of centralized system for user health tracking.

Objective of the System

General Objective

To develop a Health Tracker System that helps users monitor and manage their health information efficiently using a computerized system.

Specific Objective

1. To create a user registration and login system.
2. To develop a BMI calculation feature.
3. To allow users to record symptoms and daily health assessments.
4. To store health records in a relational database.
5. To display health history using DataGridView.
6. To implement CRUD operations using ADO.NET.
7. To ensure data validation and secure database interaction.

Scope and Delimitation

Scope

The system focuses on the following functionalities:

- User Registration
- User Login Authentication
- BMI Calculation
- Symptoms Recording
- Daily Health Assessment
- Viewing Health History
- Database Storage using SQL Server LocalDB
- CRUD Operations

Limitations

- The system only operates on Windows devices.
- Internet connectivity is not required.
- The system is intended for educational purposes only.
- The system does not provide medical diagnosis.
- Only authorized users can access their health information.

Significant of the Study

Students

The system serves as a learning tool for students in understanding software development, database integration, and health monitoring systems.

Users

The application provides users with an organized and efficient way to monitor their health data.

Researchers

The study may serve as a reference for future researchers developing healthcare-related systems.

Sustainable Development Goal (SDG)

The Health Tracker System promotes awareness of personal health conditions through digital monitoring and health record management.

II. SYSTEM ANALYSIS

Existing System

Tracking Health (e.g., hours of sleep, how many water intake, etc..) are usually done through paper-base or memory-based tracking. These methods often lead to many issues such as inaccurate data recording, misplacing the record, and difficulty in organizing records and inability to monitor historical data efficiently.

Proposed System

The proposed Health Tracker System automates health monitoring processes by providing computerized recording, BMI computation, health history management, and secure user authentication.

Functional Requirements

The system shall:

1. Allow users to register accounts.
2. Allow users to log in securely.
3. Calculate BMI automatically.
4. Record symptoms and daily health assessments.
5. Save data into SQL Server LocalDB.
6. Display health history records.
7. Allow CRUD operations.
8. Validate user inputs.

Non-Functional Requirements

- **Performance**
  The system must respond quickly to user inputs and database transactions.
- **Security**
  The system uses parameterized queries to prevent SQL injection.
- **Reliability**
  The system ensures stable database connectivity and proper error handling.
- **Usability**
  The system provides a user-friendly graphical interface.
- **Maintainability**
  The system uses modular programming for easier maintenance.

III. SYSTEM DESIGN

System Architecture

The system uses a three-layer architecture:

1. Presentation Layer – VB.NET Windows Forms
2. Business Logic Layer – Validation and Processing
3. Data Layer – SQL Server LocalDB using ADO.NET

Database Design

The system uses Microsoft SQL Server LocalDB integrated through ADO.NET SqlClient.

Database Name

- HealthTrackingDB.mdf
- HealthTrackingDatabase.mdf

Entity Relationship Description

Table: Users

Field Name	Data Type	Key	Constraints	Description
user_id	INT	PK	IDENTITY(1,1), NOT NULL	Unique user ID
full_name	NVARCHAR(255)		NOT NULL	Full name of user
username	NVARCHAR(255)		UNIQUE, NOT NULL	Username for login
password_hash	VARCHAR(255)		NOT NULL	Encrypted password
salt	NVARCHAR(255)		NOT NULL	Randomly generated unique string used to protect the password hash.
role	NVARCHAR(50)		DEFAULT 'USER', NOT NULL	User role

Table: health_records

Field Name	Data Type	Key	Constraints	Description
record_id	INT	PK	IDENTITY(1,1), NOT NULL, CLUSTERED	Unique record ID
user_id	INT	FK	REFERENCES users(user_id), NOT NULL	Connected user
bmi	FLOAT (53)		NOT NULL	Calculated Body Mass Index numerical value.
symptoms	NVARCHAR(255)		NOT NULL	Text description of symptoms logged by the user.
assessment	NVARCHAR(255)		NOT NULL	Generated system

				analysis or advice based on metrics.
created_at	DATETIME		DEFAULT GETDATE(), NOT NULL	Timestamp when the health entry was submitted.

Table: system_logs

Field Name	Data Type	Key	Constraints	Description
log_id	INT	PK	IDENTITY(1,1), NOT NULL, CLUSTERED	Unique log ID
user_id	INT	FK	REFERENCES users(user_id), NOT NULL	Connected user tracking their action
action	NVARCHAR(100)		NOT NULL	System activity (login, log out, etc.)
log_time	DATETIME		DEFAULT GETDATE(), NOT NULL	Login datetime

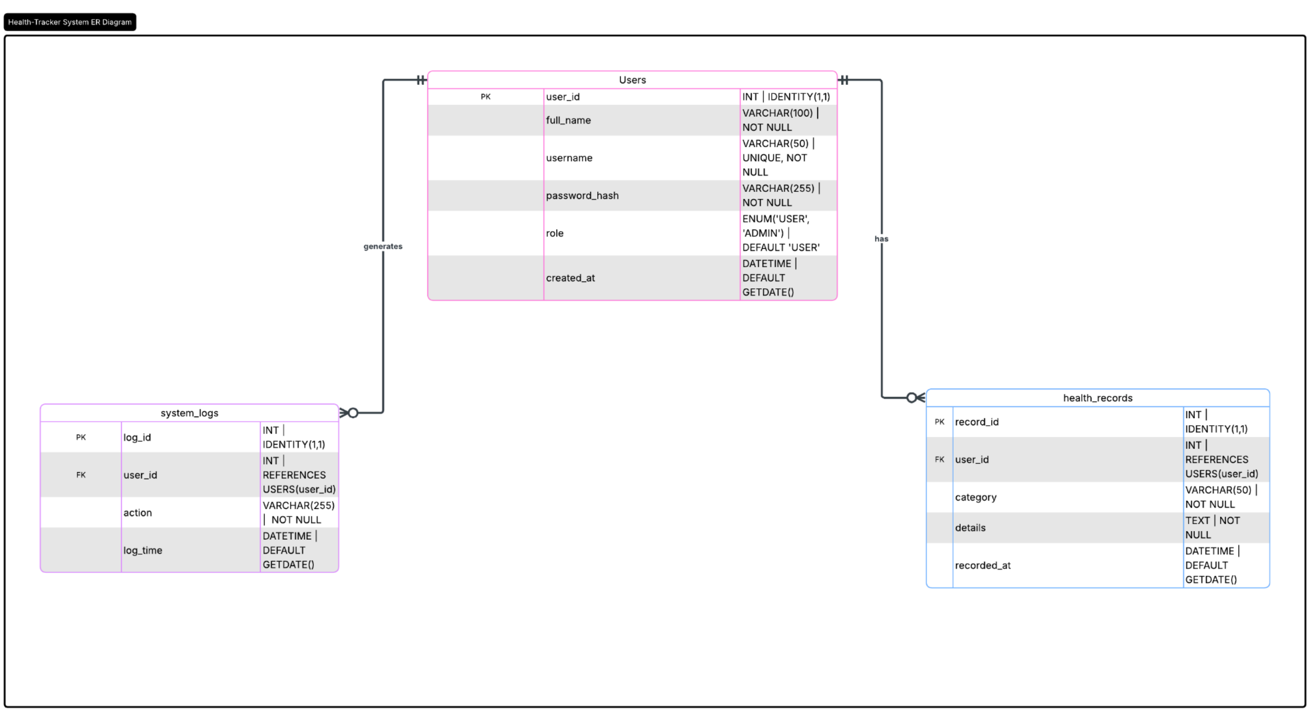
Relationship

The Users table is connected to the HealthRecords table using UserID as a foreign key.

Relationship:

Users.UserID → HealthRecords.UserID

Database ERD



DFD Diagram

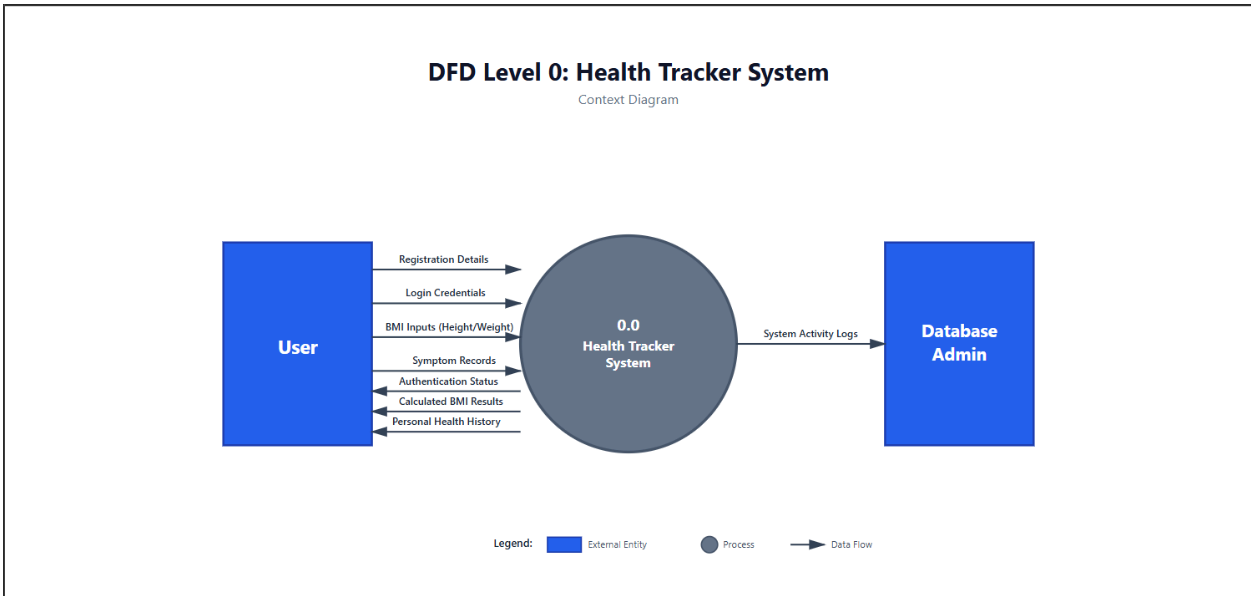
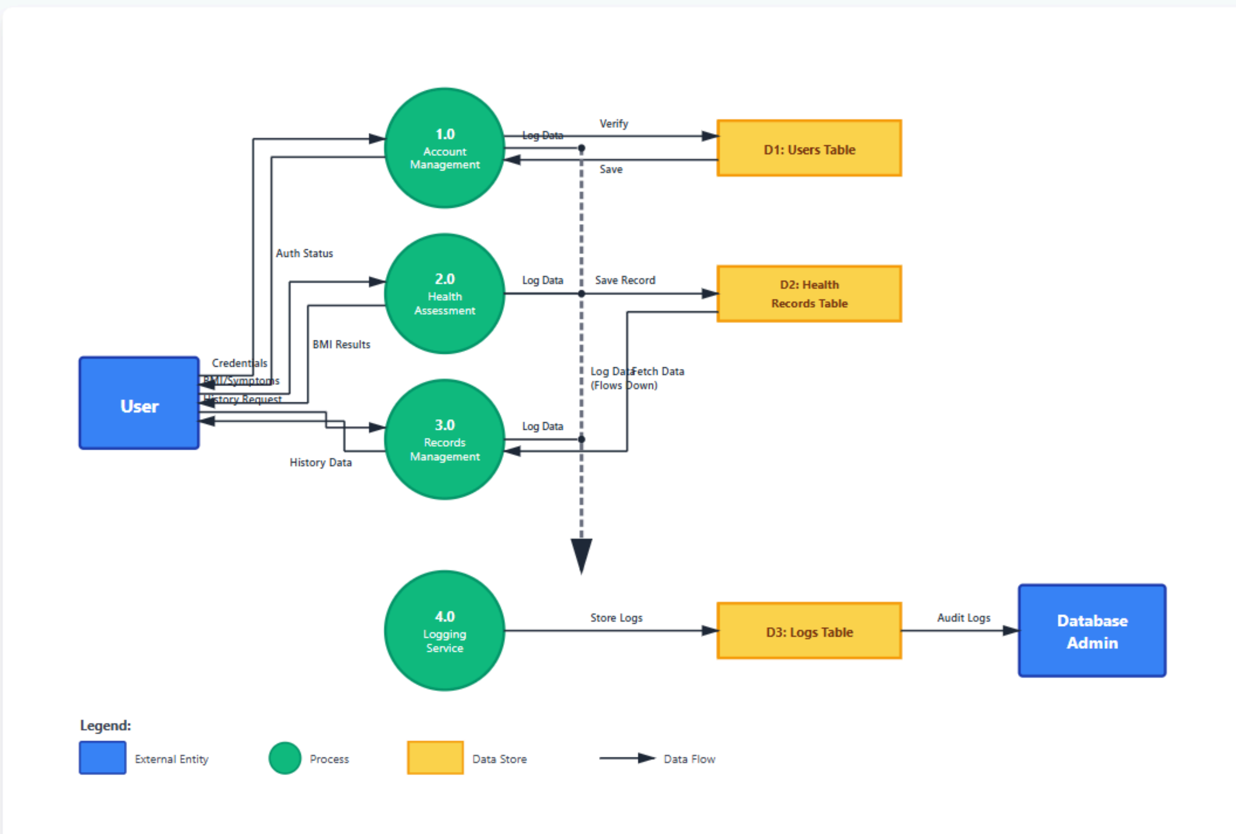


Figure 4.2: Health Tracker System – DFD Level 1 Decomposition Diagram



MANAGE RELATIONSHIPS EXPLANATIONS

1. USERS → HEALTH_RECORDS (One-to-Many):

- This relationship outlines the allocation of health records to each profile.

- **Logic:** A user can take multiple assessments over different points in time, whether it be a BMI check, a symptom check or daily wellness log, but a specific record can only belong to one user.

- **Primary Key (PK):** user_id in the USERS table.

- **Foreign Key (FK):** user_id in the HEALTH_RECORDS table.

- **Visual Indicator**: The line connecting the two tables has a single dash (One) on the USERS table side and a crow's foot (Many) on the HEALTH_RECORDS table side.

2. USERS → SYSTEM_LOGS (One-to-Many):

This connection keeps a record of what users have done on the system over time.

Reason: Since users can have a lot of logs made for them such as, if they log in, register, or do something else, but each of these logs can be traced back to the specific person who did the thing.

Primary Key PK: user_id in USERS.

Foreign Key FK: user_id in SYSTEM_LOGS.

Line indicator: In the ERD, you'll see a single dash (One) on the USERS side and a crow's foot (Many) on the SYSTEM_LOGS side.

SQL Scripts:

```
-- ==========================================
-- HEALTH-TRACKER SYSTEM DATABASE
-- ==========================================

CREATE DATABASE HealthTrackerDataBase;
GO

USE HealthTrackerDataBase;
GO

-- ==========================================
```

```
-- TABLE: Users (With Salt Column)
-- ==========================================
CREATE TABLE [dbo].[users] (
   [user_id]       INT           IDENTITY(1,1) NOT NULL,
   [full_name]     NVARCHAR(255)  NOT NULL,
   [username]      NVARCHAR(255)  NOT NULL UNIQUE,
   [password_hash] NVARCHAR(255)  NOT NULL,
   [salt]          NVARCHAR(255)  NOT NULL, -- Added for extra password security
   [role]          NVARCHAR(50)   DEFAULT 'USER' NOT NULL,
   [created_at]    DATETIME       DEFAULT GETDATE() NOT NULL,
   PRIMARY KEY CLUSTERED ([user_id] ASC)
);
GO

-- ==========================================
-- TABLE: Health Records
-- ==========================================
CREATE TABLE [dbo].[health_records] (
   [record_id]  INT           IDENTITY(1,1) NOT NULL,
   [user_id]    INT           NOT NULL,
   [bmi]        FLOAT(53)         NOT NULL,
   [symptoms]   NVARCHAR(255)  NOT NULL,
   [assessment] NVARCHAR(255)  NOT NULL,
   [created_at] DATETIME       DEFAULT GETDATE() NOT NULL,
   PRIMARY KEY CLUSTERED ([record_id] ASC),
   CONSTRAINT [fk_healthrecords_user]
      FOREIGN KEY ([user_id]) REFERENCES [dbo].[users] ([user_id])

);
GO

-- ==========================================
-- TABLE: Logs
-- ==========================================
CREATE TABLE [dbo].[logs] (
   [log_id]   INT           IDENTITY (1, 1) NOT NULL,
   [user_id]  INT           NOT NULL,
   [action]   NVARCHAR (100) NOT NULL,
   [log_time] DATETIME       DEFAULT (getdate()) NOT NULL,
   PRIMARY KEY CLUSTERED ([log_id] ASC),
      FOREIGN KEY ([user_id]) REFERENCES [dbo].[users] ([user_id])
);
GO
```

IV. SYSTEM IMPLEMENTATION

Programming Language and Tools

| Tool | Purpose | |---|---|---| | VB.NET | Application Development | | Visual Studio Community | Development Environment | | SQL Server LocalDB | Database Management | | ADO.NET SqlClient | Database Connectivity | | Windows Forms | Graphical User Interface |

Database Connectivity

The system uses ADO.NET SqlClient to establish communication between the VB.NET Windows Forms Application and SQL Server LocalDB. The database connection is handled inside the HealthLogic class which centralizes reusable database functions and logic operations.

Example:

```
Imports System.Data.SqlClient

Public Class HealthLogic

    Public Shared conn As New SqlConnection(
        "Data Source=(LocalDB)\MSSQLLocalDB;" &
        "AttachDbFilename=|DataDirectory|\HealthTrackerDataBase.mdf;" &
        "Integrated Security=True")

End Class
```

CRUD Operations

The system implements full CRUD (Create, Read, Update, Delete) operations using parameterized SQL queries to ensure data security and prevent SQL Injection attacks.

Create

Allows users to:

- Register accounts
- Save BMI results
- Store daily health assessment records
- Insert symptom tracking information into the database

Read

Allows retrieval and display of:

- User health history
- BMI records

- Assessment data
- Stored health information using DataGridView

Update

Allows modification of:

- Existing health records
- User information
- Stored assessment details

Delete

Allows removal of:

- Selected health records
- Existing saved entries from the database

Validation

The system implements input validation to ensure data integrity.

Validation Features

1. Required field validation
2. Numeric validation
3. Empty field checking
4. Error message prompts

Example:

```
If txtWeight.Text = "" Then
    MessageBox.Show("Weight is required")
    Exit Sub
End If
```

V. TESTING AND EVALUATION

Testing Procedure

The system was tested to verify:

- Database connectivity
- CRUD functionality
- User interaction
- Input validation
- Form navigation
- Data persistence

Test Cases

Test Case	Expected Result	Status
User Registration	User data saved successfully	Passed
User Log-In	Valid login access	Passed
BMI Calculation	Correct BMI output	Passed
Save Health Record	Record inserted successfully	Passed
Display History	Display displayed in DataGridView	Passed

Evaluation

The Health Tracker System successfully met the major project requirements including:

- VB.NET Windows Forms Application
- ADO.NET Database Integration
- SQL Server LocalDB
- Create, Read, and Update Database Operations
- Input Validation
- Multiple Forms Integration
- Data Persistence
- Object-Oriented Programming using Classes
- Modularized Code Structure

The project provides an organized digital solution for monitoring and storing health-related information

VI. SYSTEM FEATURES DESCRIPTION

User Registration

The system allows users to create an account by entering their personal information. Registered users can securely access the Health Tracker System using their login credentials. The registration form validates user input before storing the information in the database.

Features:

- User account creation
- Input validation
- Database record insertion
- User information storage

User Login

The Login Form allows registered users to access the system by entering valid credentials. The system checks the entered username and password against the records stored in the SQL Server LocalDB database.

Features:

- Username and password authentication
- Validation of login credentials
- Access to Dashboard upon successful login
- MessageBox prompts for invalid login attempts

Dashboard

The Dashboard serves as the main navigation area of the application. It allows users to access different health monitoring features available in the system.

Features:

- Navigation to BMI Calculator
- Navigation to Daily Assessment
- Navigation to Symptoms Monitoring
- Navigation to Health History records

BMI Calculator

The BMI Form allows users to calculate their Body Mass Index (BMI) using height and weight inputs. The calculated BMI result helps users monitor their health condition.

Features:

- BMI calculation
- Weight and height input validation
- Health status display
- Saving BMI results into the database

Daily Health Assessment

The Daily Assessment Form allows users to input and store their daily health condition and monitoring information.

Features:

- Daily assessment entry
- Health monitoring
- Data storage using SQL Server LocalDB
- User-friendly form interface

Symptoms Monitoring

The Symptoms Form allows users to record symptoms related to their health condition. This feature helps users monitor health concerns regularly.

Features:

- Symptom recording
- Health information management
- Data persistence
- Organized symptom tracking

Health History

The History Form displays previously saved health records and assessment information.

Features:

- Viewing stored records
- Displaying data using DataGridView
- Retrieval of saved database information
- Health monitoring history

VII. INDIVIDUAL CONTRIBUTIONS

Members	**Contributions**
Arcales	Integration & ADO.NET Developer
De Guia	Database Engineer
Diaz	Backend / Logic Developer Conversion of the code from previous project DSA (C++ -> VB.Net)
Kismundo	Documentation & Reports
Marasigan	UI Developer (Forms & Design)

VIII. CONCLUSIONS

The Health Tracker System offers a modern solution that allows users to monitor their health.

The Health Tracker System:

- Organizes data more effectively than previous approaches
- Provides an automated method for calculating BMI
- Supports efficient storage and retrieval of records using SQL Server LocalDB with ADO.NET as the method to integrate the two systems.

This project demonstrates the following practical uses:

- Object-Oriented Programming (OOP)
- Database Management
- Windows Forms Application Development
- Validation Techniques

The Health Tracker System also helps meet United Nations Sustainable Development Goal 3 of promoting good health through health awareness, and assisting people in being able to keep organized digital medical records.

IX. REFERENCE

- Microsoft Documentation – VB.NET
- Microsoft Documentation – SQL Server
- Microsoft Documentation – ADO.NET SqlClient
- Sustainable Development Goals (SDG 3)

X. APPENDICES

Appendix A – Existing Project Files

Forms

- BMIForm.vb
- DailyAssessmentForm.vb
- Dashboard.vb
- HistoryForm.vb
- LoginForm.vb
- RegisterForm.vb
- SymptomsForm.vb

Logic File

- HealthLogic.vb

Configuration and Database Files

- App.config
- HealthTrackerDataBase.mdf
- HealthTrackerDataBase.ldf

Appendix B – Database Files

The system uses SQL Server LocalDB database files:

- HealthTrackerDataBase.mdf
- HealthTrackerDataBase.ldf

These database files are connected to the VB.NET Windows Forms Application using ADO.NET SqlClient to support persistent data storage and retrieval of health records.

Appendix C – Technologies Used

Technology	**Purpose**
VB.NET	Application development
Windows Forms	User Interface development
ADO.NET SqlClient	Database connectivity
SQL Server LocalDB	Data storage

Visual Studio Community	Development environment

Appendix D – Sustainable Development Goal Alignment

The project supports:

SDG 3 – Good Health and Well-Being

The Health Tracker System helps users monitor and manage their health records digitally. The system promotes awareness of personal health conditions through BMI calculation, symptom monitoring, and daily health assessments.